# Nguyễn Quốc Tuấn

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## Introduction

### AI Engineer - Computer Vision

AI Engineer with 1+ year of production experience building end-to-end Computer Vision systems, from model training to real-time edge deployment. Focused on turning slow, manual, or unsafe workflows into fast, automated processes. Experienced on fusing/quantizing model for edge deployment.

**Code:** github.com/tuan2k33 | **Portfolio:** tuan2k33.github.io

## Experience

**Vinsmart Future** Ho Chi Minh City
*AI Engineer* *Aug 2026 – Sep 2026*

***Truck tarp Violation Detection*** | Python, DETR, ResNet
- Built a full violation detection pipeline to flag uncovered construction-material trucks.
- Achieved 91% end-to-end accuracy, 97% recall.

***Household Re-ID system for Smart-home*** | Python, ViT, PyTorch
- Developed a system assigning IDs to residents, neighbors, and pets to flag unknown persons or animals, finetuning a ViT backbone with a custom retrieval head.
- Reached 95% mAP and 99% Rank-1 accuracy.

**Rainscales Viet Nam** Ho Chi Minh City
*AI Engineer* *Jun 2025 – Jun 2026*

***Warehouse Inventory Counting System***| Python, ResNet, YOLO, OCR, decord, OpenCV
- AI camera pipeline replacing manual pallet counting, improved worker safety.
- 5x speedup, 90-94% bin-extraction accuracy, 81% accuracy on box count/fullness/SKU.

***Event Analytics Pipeline*** | Python, NVIDIA DeepStream, ServiceMaker
- Deployed real-time video analytics on Jetson, handling multiple RTSP streams with a store-and-process pipeline (MP4/JSON chunking, rule engine, rolling buffering).
- Scaled to 8 cameras at 25fps on Jetson Orin Nano and 60 cameras at 25fps on Jetson AGX Orin.

***Model Research & PoC*** | Python, YOLO, ResNet, TensorRT
- Trained a warehouse object detector with 3M images, 16 classes, achieve 85% mAP.
- Trained a PPE compliance classifier (helmet/vest/coverall), achieve 88% accuracy, false alarm 5%.
- Fused and quantized models into lightweight TensorRT engines for real-time inference on Jetson.

## Education

**Ho Chi Minh City University of Technology (HCMUT – VNUHCM)** Ho Chi Minh City
*Bachelor of Science – Computer Science* *2026*

## Research

***SMC2L: Self-Supervised Mixup Contrastive Continual Learning*** | Thesis | Paper
- Graduation thesis proposing a self-supervised continual learning method (contrastive loss + pixel mixup + distillation) to reduce catastrophic forgetting.
- 92% accuracy on satellite dataset; accepted at IAAA 2026 (Springer LNNS, SCOPUS-indexed).

***SAM3-DeepStream*** | Python, C++, SAM3, TensorRT, NVIDIA DeepStream
- Personal research specializing Meta's SAM3 (open-vocabulary detection + segmentation) into NVIDIA DeepStream's nvinfer – exporting a TensorRT engine with a custom C++ mask/bbox parser, supporting on-the-fly model updates.
- Reached real-time throughput on 4-camera RTSP.

## Skills

**Coding:** Python, C/C++, R, MATLAB
**Machine Learning:** PyTorch, OpenCV, YOLO, Resnet, OCR
**MLOps & Edge:** TensorRT, NVIDIA DeepStream, Triton, Docker, Linux (Ubuntu)
**Database:** SQL (SSMS, MySQL)
**Web Dev:** PHP, HTML, CSS, JavaScript

## Certifications

**IELTS Band 6.5**
**HackerRank SQL (Advanced)**